Undo and End Process

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This document describes the Undo Order/Wave, End Work for a Wave and Close Order process and its associated updates.

1. Undo Wave/Order - The process that brings the order back to its original state. It brings back the orders that are allocated to released status.
   - Undo Order action from Order UI
   - Undo Wave action from Wave Runs UI
   - The order is deselected because of failure during processing at various stages.
2. End Work - The process that cancels the un-executed work (oLPNs/Task Details).
   - End Work action from Wave Runs UI

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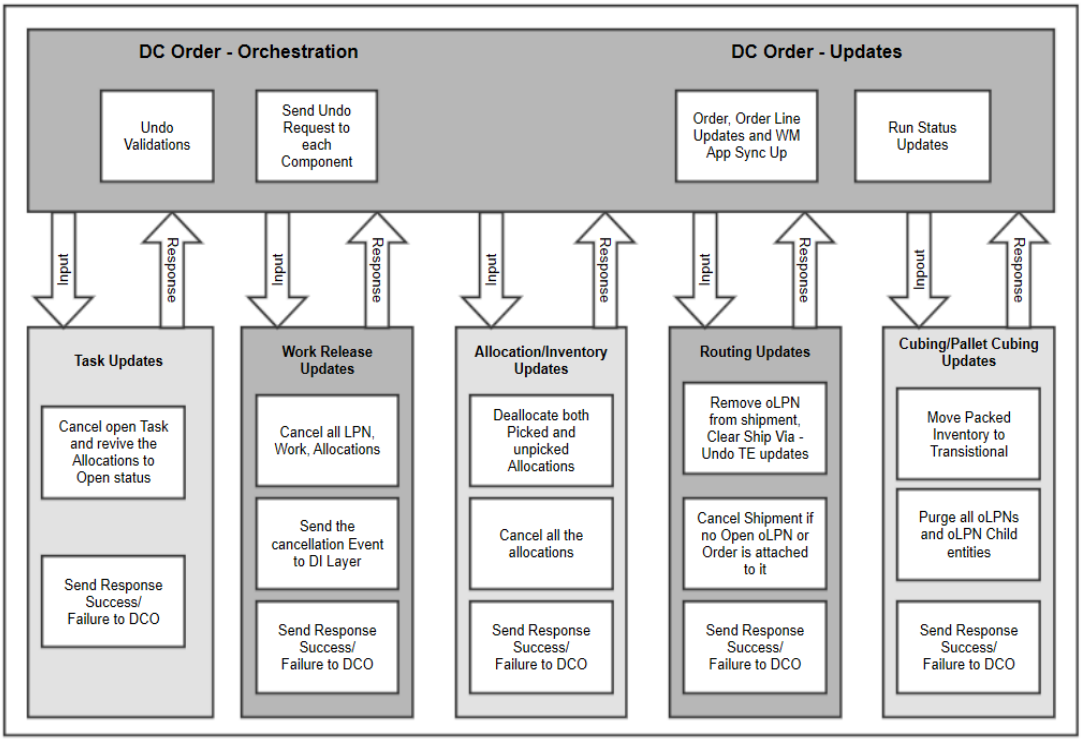
3. Close Order - The process that cancels the un-allocated order lines.
    - Close Order action from Orders UI

Process Flow



Undo Process

The Undo process can be triggered manually from the Order UI or Wave Runs UI. The DC Order process orchestrates the Undo process. The Undo process is triggered when there is a failure during order processing at various steps.

For example, when an order fails cubing, or fails to find a consolidation location, the order is deselected. This triggers the Undo Order process. During order deselection, where an Order Line fails during processing, the Order Lines in the run are deselected to allow the rest of planning run to progress ahead in the pipeline flow. This can happen during a Wave or Stream run. DC Order receives a request to orchestrate the deselection process for the Order Lines as part of the run.

Un Plan Order Schedule scheduler evaluates Undo or Un-plan Order requests with status PENDING_Pipeline_cancelation. It evaluates the pipeline status of the lines for each order, to check if they have received pipeline response. The requests are accordingly moved to an error state if necessary; otherwise, they are moved to a completed state and the un-plan/Undo process begins.

Undo Validations

DC Order performs validation, before an Undo Order request can be orchestrated. The below process flow diagram highlights the different validations:

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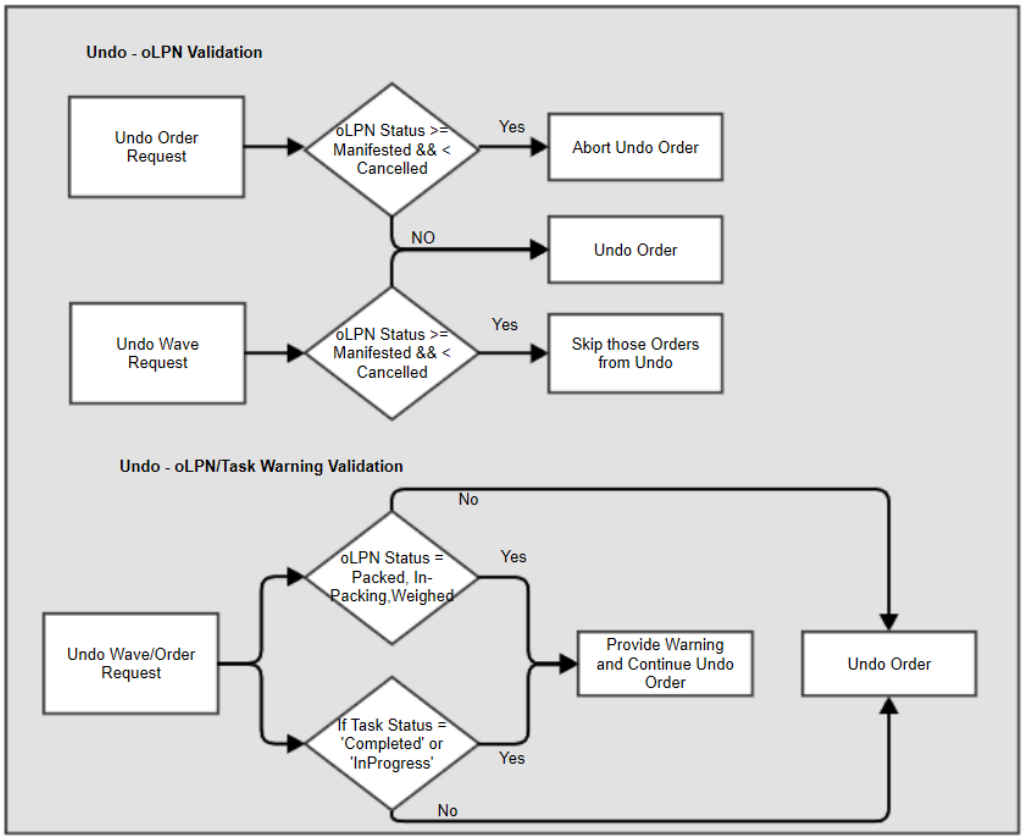

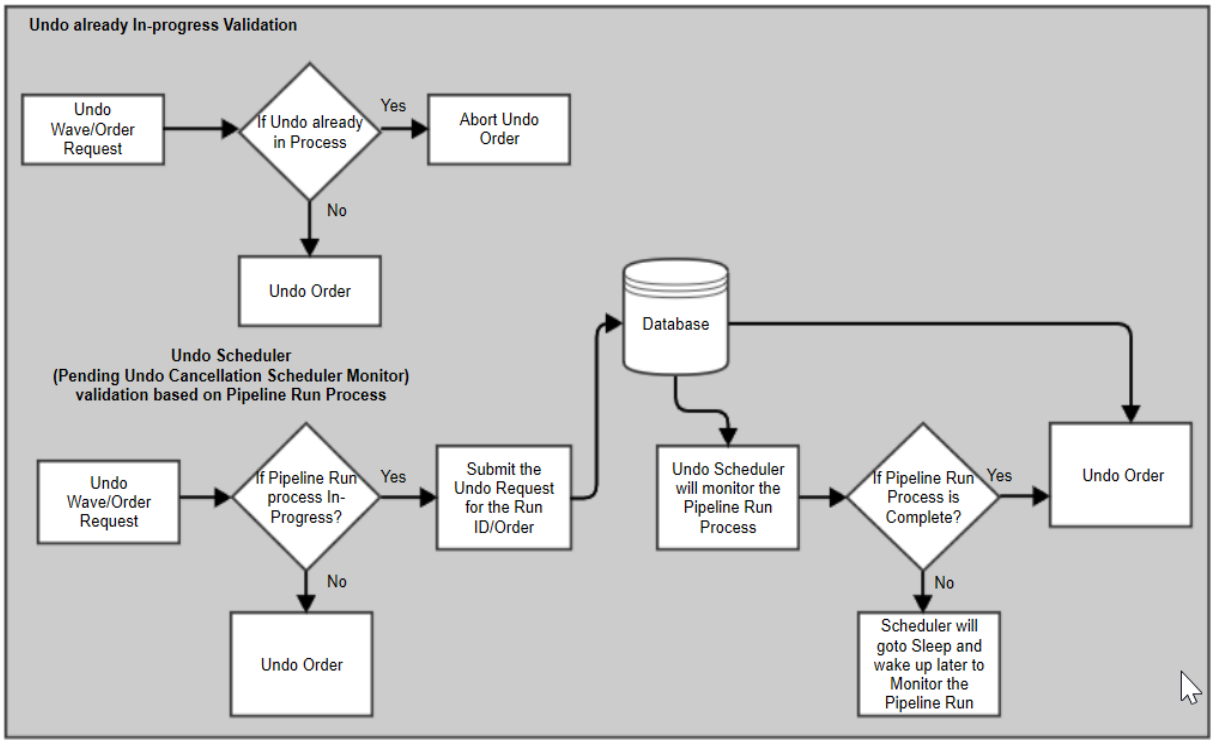

Order UI Validations

- Error response if any of the oLPNs for an order is already manifested, loaded, or invoiced (oLPN status >= 'Manifested')
- Error response if an Undo process for the same order is already in progress.
- Warning response if the user is trying to undo an order which is already in the pipeline process (Wave or Stream run process). In

this scenario, DC Order provides a warning, and accepts the undo request for the order, triggering 'Undo' only after the pipeline run process is complete. This Undo request is stored in a new entity and later processed via a scheduler (Pending Undo cancelation Scheduler Monitor) once the pipeline process is complete.

- Warning response if any of the allocations for the order is in-progress or complete, or if any of the oLPNs for the Order is in-packing or packed status (Task Status >= 'In-Progress' or oLPN Status >= 'In-Packing' and oLPN Status <= 'Weighed').

Wave Runs UI Validations

- Error response if the user is triggering 'Undo' for a Wave for which undo is already in progress.
- Warning Response if any of the allocations for the order is in-progress, or complete, or if any of the oLPNs for the order is in-packing or packed status (Task Status >= 'In-Progress' or oLPN Status >= 'In-Packing'). When Undo Wave is triggered, this process skips the orders for which any oLPN is in status >= 'Manifested' status.
- Warning Response if the user is trying to undo a Wave which is already in the pipeline process (Wave run process). In this scenario, DC Order provides a warning, and accepts the Undo request for the order, triggering 'Undo' only after the pipeline run process is complete. This Undo request is stored in a new entity and later processed via a scheduler (Pending Undo cancelation Scheduler Monitor), once the pipeline process is complete.

Undo Validation Updates

Undo is orchestrated by DCO and DCO makes component calls to revert the order planning changes.

Undo Process Sequence

The Undo Process is performed in a specific static manner to ensure updates take place in a specific sequence. The sequence is as follows:

1. Task Component: Cancel all open pick tasks
2. Custom Process: Perform any custom Undo updates needed
3. Work Release: Cancel work/allocations
4. Allocation: Clear the allocated Inventory
5. Routing: Undo Routing updates
6. Cubing/Pallet Cubing: Cancel oLPN updates and delete pallet records if palletized.
7. DC Order: Perform Order and Order Line updates and Undo Run updates

Each process receives an input from DC Order with the below input for performing the Undo process.

Process	Input
Undo Order	Order ID
Undo Wave	Order ID + Run ID
Deselect Order	Order ID + Run ID

Undo Process

Task

In the case of Undo Wave, the task component cancels all the open task details for the Run ID. For Undo Order, it instead cancels all open task details for the Order ID. Once all the task details are canceled for a task, the task header is also canceled. Once the updates are performed, a success message is sent to DC Order to orchestrate the next step for the Undo process.

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Custom

A Custom process in the pipeline can perform Order updates. A user exit is available to implement required custom 'Undo' updates.

Work Release

The Work Release process cancels all associated allocations in Work Release. Allocation cancelation is rolled up to Work and Task records; if all the allocations of work/task are canceled, the work and/or task are canceled as well. Work Release also sends an event to EIF (Equipment Integration Framework) to enable sending of cancelation messages to external Warehouse Control Systems. For example, all allocation and oLPN divert messages would likely need to be canceled via a cancelation message sent to Picking and Sorting systems, respectively. This process is triggered only in the case of Undo Order and Undo Wave.

Allocation

The Allocation process clears allocated inventory for any open allocations. This clears LPN, or any allocated inventory from pick locations. Once the allocation is cleared, the allocations are canceled. This performs updates to the iLPN allocated quantity, status, and pick location To Be Picked quantities. De-Allocation PIXes are generated for the de-allocated quantity.
In case of picked inventory, where the pick allocation is complete and the pack allocation is created but not yet completed, pack allocations are de-allocated and canceled. The Tote ID or LPN ID that contains the picked inventory is also de-allocated and updated to 'In Inventory Not Putaway' status. Any replenishment request raised for the Order Lines during allocation process are also canceled.

Routing

The Routing process performs Undo Routing updates by removing the oLPN/Order from its planned shipment or removing the ship via from the oLPN/Order. Undo Routing updates are performed if the routing is both part of the pipeline and configured to occur after the allocation step in the pipeline. If routing is performed as standalone, or the routing step is configured to occur before allocation step, then Undo Routing updates are not performed for those orders.

Cubing

The Cubing process cancels all open tasks for that oLPN, such as loading tasks for the oLPN. It performs de-consolidation updates for the oLPN, if the oLPN was originally consolidated. This removes the destination location from the oLPN, decrements the directed quantity, and removes any locks on the consolidation location if necessary. If the oLPN was packed fully or partially, then it de-allocates packed inventory. This reduces the inventory from oLPN and transfers the inventory into an iLPN, or into a Lost inventory bucket, per configuration. Once these updates are performed, the oLPN details and oLPN header are canceled.

Pallet Cubing

When all the oLPNs on the pallet are canceled or deleted, the corresponding Pallet record is also deleted.

Order Consolidation

Undo of consolidation is performed by the Undo Cubing process anytime, an oLPN is canceled or purged (Undo flow), as mentioned above. The consolidation updates are reverted just prior to the oLPN status updates.

DC Order

The DC Order process decrements the allocated quantity and user canceled quantity from the Order Lines. It also removes the Order Planning Run ID from the Order Lines. The statuses for the Order and Order Lines are updated based on the allocated quantity. Whenever there is a partial allocation during the allocation process, the Order Lines are split into multiple child Order Lines containing the unallocated quantity of the Order Line. The Undo process do not merge the child Order Lines into parent Order Lines. An Order status change PIX message, is also generated in the case of Order status update. DC Order generates the de-allocate Order PIX message, to indicate the host, that the Order Line is de-allocated. In the case of deselecting Order or Order Line

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process within a run due to a failure, deselected Order and Order Lines are updated in the Order Run Summary entity. Order Planning Run Status is also updated to 'Canceled' if all the Orders and Order Lines in the run are unplanned through Undo Order.
The unplan Runcancelation MonitorSchedule scheduler evaluates un-plan (Undo) run requests that are not in a completed or error state. It checks for Order Plan Run completion and Undo Run completion. If no Order Lines can be found with a Planning Run ID matching the Undo Run request(s), it updates the Order Planning Run to CANCELED. If all Undo Order requests that are part of a Planning Run are complete, it updates the Undo Run request to completed. If all are completed but some have errors, it updates the Undo Run to partially completed.

Chase Process

Undo of Chase Wave/Original Wave is not supported for orders containing oLPN Level Need. The reasons are:

- In scenarios wherein an oLPN is packed, when Undo is initiated, it could take the oLPN back to an In-Packing status, which is operationally not a valid behavior.
- Modifying oLPNs that are partially shorted, with remaining portion being processed in the floor, needs to be avoided as much as possible.
- If oLPN is fully shorted and chased, cancel oLPN can be used instead of Undo.

User can also use Pick Shorting/Undo Order (if not already manifested) to Undo Orders containing oLPN level Need. Undo of Chase Wave/Original Wave is done for orders containing Order Level Need.

Updates DC Order

Entity	Attribute	Comments
DCO_ORDER_LINE	ALLOCATED_QUANTITY	set to 0.0
DCO_ORDER_LINE	ADJUSTED_ORDERED_QUANTITY	value of ORDERED_QUANTITY
DCO_ORDER_LINE	ORDER_PLANNING_RUN_ID	set to null
DCO_ORDER_LINE	ORDERED_QUANTITY	update to original Ordered qty before the line was split in case of shortage
DCO_ORDER_LINE	PENDING_REPLENISHMENT	Set to FALSE
DCO_ORDER_LINE	NBR_OF_ALLOCATION_ATTEMPTS	set to 0
DCO_ORDER_LINE	Pipeline_STATUS	set to 1000

De-allocation order PIX event is generated for the allocated quantity that is decremented for the order line. The PIX generated has to be for the original order.

Allocation

All open allocations will be canceled. This cancelation includes pick pack and pack allocations. De-allocation inventory PIX event is generated for the allocations that are cancelled.

Cubing

Upon Undo initiation, the Cubing component reverses the changes done during chase wave process.

Consolidation

Undo Consolidation is performed if either the oLPN has a destination or if the execution is complete and the oLPN is already located.

oLPN destination

Entity	Attribute	Comments
LPN	Destination Sub Location ID	Destination Location ID is cleared out.

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Entity	Attribute	Comments
PKT_CONSOL_LOCN	Directed UOM Quantity	Directed UOM quantity is reduced by 1.
PKT_CONSOL_LOCN	Directed Weight	Directed Weight reduced by the oLPN estimated Weight.
PKT_CONSOL_LOCN	Directed Volume	Directed Volume reduced by the oLPN estimated Volume.

Execution is complete and oLPN is located

Entity	Attribute	Comments
LPN	Current Sub Location ID	Current Location ID cleared out.
LPN	Staging Indicator	Staging Indicator updated to '0'
PKT_CONSOL_LOCN	Current UOM quantity	Current UOM quantity is reduced by 1.
PKT_CONSOL_LOCN	Current Weight	Current Weight reduced by the oLPN estimated Weight.
PKT_CONSOL_LOCN	Current Volume	Current Volume reduced by the oLPN estimated Volume.

Routing

On Undo, Routing component reverses all updates that were done as part of the routing process.

These include the following:

- LPN Updates - Nullification of Ship Via, tracking number, charges (Base charge, Additional option charge, Actual cost), Calculated Cut-off Time, LPN-Level Routing flag; removal of LPNs from the Shipment (for non-parcel), Routing/Rating lane and lane detail IDs
- Order Updates - Nullification of Line Haul Ship Via "Is Routed" flag, Shipment ID, Assigned Mode, Assigned Service Level and Assigned Ship Via

On successful completion of Undo, routing component synchronizes the update to the DCO component. In an event where Undo process fails, the error description and the error code are updated to the DCO component.

Work Release

Entity	Attribute	Comments
WR_ALLOCATION	STATUS	Status is set to 1300.00 - canceled.
WR_WORK	STATUS	Status is set to 1300.00 - canceled.
WR_TASK	STATUS	Status is set to 1300.00 - canceled.

Chase

Scenario 1: Wave-processed orders which contain shortages not yet chased. Undo the Wave.

Scenario conditions: Orders exist with DCO_ORDER_LINE.SHORTAGE_LINE = 1, and all such lines in released or canceled status.

Entity	Attribute	Comments
DCO_ORDER_LINE_SHORTAGE	NA	Entry deleted based on Order_LINE_PK
DCO_ORDER_LINE	NA	Entry deleted where DCO_ORDER_LINE.SHORTAGE_LINE = 1
PICKING_SHORT_ITEM	NA	Entry deleted corresponding to DCO_ORDER_LINE_SHORTAGE

Scenario 2: Wave-processed orders contain shortages that have been chased. Undoing one of the corresponding chase waves, acting only on shortage lines

Scenario conditions: Some orders in the wave have lines with DCO_ORDER_LINE.SHORTAGE_LINE = 1 and several such lines in > released status and < canceled status, with wave number populated.

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Entity	Attribute	Comments
DCO_ORDER_LINE_SHORTAGE	NA	Entry deleted based on Order_LINE_PK
DCO_ORDER_LINE	FULFILLMENT_STATUS	Set to RELEASED for DCO_ORDER_LINE.SHORTAGE_LINE = 1 so this can be waved again (other Order Line updates similar to DC Order updates above)
PICKING_SHORT_ITEM	STATUS	Set to unprocessed

Scenario 3: Wave-processed orders contain shortages that have been chased. Undoing original wave, acting on both shortage lines and non-shortage "host-sent" lines

Scenario conditions: Some orders in the wave have lines with DCO_ORDER_LINE.SHORTAGE_LINE = 1 and several such lines in > released status and < canceled status, with wave number populated:

1. If Undo of current wave sets host-sent original line to released status, DCO rolls corresponding shortage lines in released status (only Order level need) up to the original line (updates are the same as scenario 1 above).
2. Irrespective of the wave which caused the shortage, if a shortage line is in released status and the original (host-sent) line is in released status, the shortage line is merged into the original (host-sent) line (updates are the same as scenario 1 above).
    1. For example, if Wave 1 did not have a shortage, but is partially allocated; and Wave 2 had packing shortages that are not yet chased, the system does **not** merge during Undo of Wave 2, but merges during Undo of Wave 1 when the original (host-sent) line goes to released status. This occurs even though the shortage belonging to Wave 2.
3. If the shortage lines have been allocated on subsequent waves, they remain allocated even if the original (host-sent) line moves to released status. These shortage lines are not undone unless the user specifically performs 'Undo' on the waves on which the shortage lines were allocated.

End Work

End work is a process that cancels all the un-processed task for a wave. This is an action from Wave Runs UI where user will have ability to cancel any unexecuted tasks/oLPNs for the wave so that the completed oLPNs can be shipped. This process is widely used in retail waves. This action is only supported for a completed wave (single select action). DC Order triggers "End Work" from Wave Runs UI and DC Order calls Pick Pack, Work Release and DC Inventory process to perform the updates and DC Order further performs updates on the orders. On any failure, these failures are logged in order failures and failed orders are skipped from End Work process. DC Order uses order and wave as the input for this process. This process is similar to Undo process where the same orchestration model is followed for End Work process also.

Validations on End Work

1. If "End Work" action taken against a wave for which undo wave is in-progress, then error out as "Wave cannot be ended, Undo wave is in-progress".
2. End Work process skips orders from processing in the below scenarios:
    - Order is greater than packed status
    - Order submitted for Undo/Cancel process
    - Order has any oLPN level shortages (In case user has to end this work, user has to cancel the shortage or convert the oLPN level short to order level short and perform End Work again for the wave).
    - Order has an oLPN in "Picking" or "Picked" or "Packing" status.
    - Order has any PICK PACK or PACK task details in-progress
3. Wave can contain both retails, eCom and wholesale orders. So, to skip any eCom or wholesale orders from End Work processing, system uses a User Exit to perform the same. User Exit is provided to get the list of Order IDs and send back a

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response with the list of Order IDs which need to be skipped from the End Work process.

Assumption on End Work

1. PICK allocation created as part of the wave will not be canceled, these would be cancelled whenever the PACK allocation gets canceled by 'Cancel Unused Pick Allocation' process.
2. Replenishment allocation created as part of the wave will not be canceled, these would be canceled when allocated quantity gets decremented from location by 'Cancel Unused Replenishment' process.

Updates on End Work

1. Pick Pack Updates
   - Cancel all open task details (Type = PICKPACK and PACK) for the order and wave combination.
   - De-Allocate the unpicked inventory from pick locations when any PICK PACK allocation is canceled.
   - De-Allocate any picked inventory from Tote/iLPN when any PACK allocation is canceled.
   - Cancel all oLPN which are in Created status. (Perform all the cancel oLPN updates)
   - Generate cancel MHE messages for the canceled oLPNs. (Divert)
2. Work Release Updates
   - Cancel all the open work and allocation in Work Release for the order and wave combination.
   - Cancel all the work and allocations corresponding to the task details in Pick Pack (Pick Pack will trigger event to cancel these allocations in Work Release)
   - Cancel all the work and allocations corresponding to the allocation in DC Inventory (DC Inventory will trigger event to cancel these allocations in Work Release)
   - If all the allocation for a pack wave is canceled, cancel the pack wave.
   - If all the allocation in a Cart is canceled, delete the cart.
3. DC Inventory Updates
   - Cancel all the open allocation in DC Inventory for the order and wave combination.
   - De-allocate the inventory from iLPN/Location which is allocated for the order.
   - De-allocate the picked inventory based on the request from Pick Pack.
   - Generate inventory de-allocation PIX.
4. DC Order Updates
   - Cancel all the un-packed order line quantities. If no inventory is packed for the order line, cancel the order line. Generate order quantity decrease to HOST.
   - If partial quantity is packed for the order line, update the order quantity to packed quantity and generate the order quantity decrease for the quantity decreased to HOST. Order line status will get updated to Packed status.
   - Re-calculate the order status and generate order status change PIX.
   - If all the order lines are canceled, cancel the order too. Generate order status change PIX in this case.
   - Send event to routing when there is change in order quantity or status. This removes the canceled order from Shipment and updates the Shipment capacity based on the order quantity that is decremented.
   - Trigger ship confirm based on order status.

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- Update the wave status to "Ended" if all the un-executed work is cancelled for the wave (including the task details, oLPNs, allocations etc). If there is any failure for an order as part of this process, wave is not updated to "Ended" status.

Close Order

Close order is a process that cancels all the un-allocated order lines for an order. This is an action from orders UI where user has the ability to cancel any un-allocated order lines for the order. The remaining order lines which are allocated would be shipped. This process is widely used in retail waves. This action is only supported for multiple orders which has minimum status as released (multi-select action).

Validations on Close Order

1. Shortage order line will be skipped from close order process.
2. Order lines which are already allocated or selected for allocation will be skipped from close order process.

Updates on Close Order

1. Cancel all the un-allocated order line. Generate order quantity decrease to HOST.
2. Re-calculate the order status and generate order status change PIX.
3. If all the order lines are canceled, cancel the order too. Generate order status change PIX in this case.
4. Send event to routing when there is change in order quantity or status. This removes the canceled order from Shipment and updates the Shipment capacity based on the order quantity that is decremented.
5. Trigger ship confirm based on order status.

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